

- 1 (a) State Newton's law of gravitation.

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.....

..... [2]

- (b) A binary star consists of star A, of mass  $4.0 \times 10^{30} \text{ kg}$ , and star B, of mass  $2.0 \times 10^{30} \text{ kg}$ , separated by a distance of  $3.3 \times 10^{12} \text{ m}$ . The stars are both in circular orbit around their common centre of gravity X, as shown in Fig. 1.1.

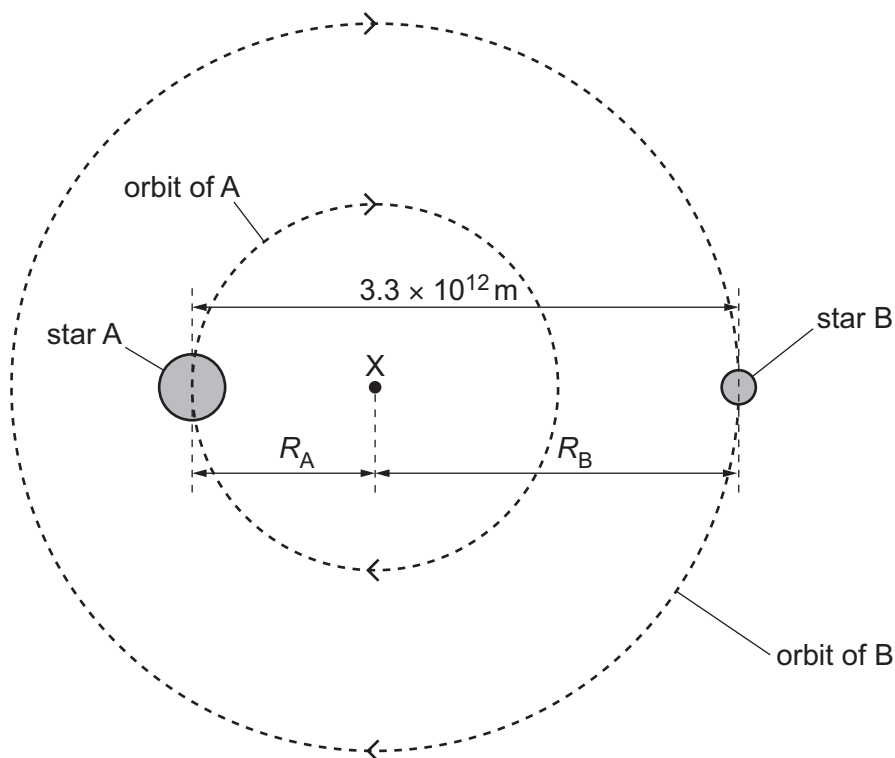


Fig. 1.1

The radius  $R_B$  of the orbit of star B is double the radius  $R_A$  of the orbit of star A.

- (i) Use Newton's law of gravitation to calculate the magnitude of the gravitational force exerted by each star on the other.

force = ..... N [2]

